

Fermenting Lagers

From German brewing and more



Primary fermentation cellar at the Paulaner Brewery, Munich, Germany

Lagers are fermented with a bottom fermenting lager yeast (*Saccaromyces pastorianus*). These yeasts are able to ferment at lower temperatures than the top fermenting ale yeasts (*Saccromyces Cervesiae*). The result of this low temperature fermentation and maturation is a beer with a cleaner flavor profile (less esters, less higher alcohols) than its top fermented counterpart. The extended cold storage (lagering) also makes these beers more shelf stable than ales which explains why most of the world's beers are of the lager variety.

Because lagers ferment at lower temperatures than ales and yeast metabolism works slower at these temperatures, they take longer to ferment and also require more attention than ale fermentations, which makes them less attractive for most home brewers. This article is aimed at the novice as well as the advanced lager brewer and tries to explain the various lager brewing techniques that home brewers use. The first section is a how-to for your first lager fermentation. The following sections delve deeper into the subject of lager fermentation and discuss the pros and cons of different fermentation techniques and also give some background information on lager fermentation in commercial breweries.

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Your First Lager Fermentation

Though many experienced brewers may read this and note that this is not the absolutely best way to ferment lagers, it is regarded as the most foolproof and that's what you are looking for for your first lager fermentation. You need the first batch to be a success to get hooked on lagers and their smooth taste. Then you may start digging deeper into this subject and find a fermentation schedule that works best for you and your set-up.

One day before brew day pitch a 2 qt (2 L) well aerated starter with an Activator Pack (Wyeast) or vial (White Labs) of the lager yeast of your choice. Both companies offer really great yeast strains. If you are looking for a versatile lager yeast go with the German Lager (WLP830 or Wyeast 2124; According to White Labs and Wyeast this is the W-34/70 strain which is the most widely used lager strain in German beers) or whatever your recipe calls for. Keep this starter at room temperature 68 - 70 °F (20 - 21 °C) and let it start fermenting. It may throw off some sulfur notes (rotten egg smell) which is common for lager yeasts.

Brew an average gravity lager OG: 1.044 - 1.056 (11 - 12 °P). These beers will not result in toxic alcohol levels for the yeast which makes for a more forgiving fermentation. Once brewed, chill the wort to a temperature below 60 °F (15 °C). The mid 50's should work best for this fermentation schedule. If you are not able to get the wort that cold with your chiller and your tap water, you can use a pump to recirculate ice water through the chiller. Because this pump doesn't have to be food grade, a simple submersible utility pump will do. Another option is to let the wort cool in your lagering fridge before pitching.

When transferring the wort into the fermenter, make sure to leave most of the hot break and hops in the kettle. This can be achieved with Whirlpooling or straining. The latter can be problematic since the fine break material tends to clog the strainer. The removal of hot break, some cold break and hops is recommended because the beer would be sitting on this trub for a long time (4 weeks) although recent studies have shown that the importance of trub removal is somewhat overstated [Kuehbeck 2007]. After transfer into the fermenter the wort needs to be aerated well. A healthy lager fermentation requires more oxygen than an ale of the same strength in order to reduce the stress on the yeast. The required oxygen level of 8-10 ppm (mg/L) is best achieved through 1 to 1.5 minutes of pure O₂ or 20-30 minutes sterile air through a 2 micron stainless steel stone.

Pitch the whole starter into the primary fermenter. Wait until you see fermentation activity (low krausen or bubbles in the airlock) until you move the fermenter to an area (basement or fridge) where you have a constant 48 - 52 °F (9 - 11 °C). Let the primary fermentation take its course for a few (3-4 weeks) until there is no airlock activity left. If you want to pitch the yeast cold (at 46 - 48 °F (8-9 °C), which is actually the preferred method, you will need to grow more yeast. This means using a starter as large as 1 gal. When pitching cold you can also expect a longer lag time. Because all these things complicate the process I'm advocating warm pitching for your first lager.

After the primary fermentation is complete, rack the beer to a lagering vessel. It can be another carboy or a soda keg with shortened dip tube. The beer is then moved to an area where the ambient temperature is between 32 and 38 °F (0 - 3 °C) where it will remain for at least another 4 weeks.

Now you can either rack to a serving keg and force carbonate, in case you didn't do the force carbonation during lagering, or bottle. If you plan to bottle condition the beer you may want to add fresh yeast with the priming sugar, because the yeast present in the beer may not perform as well anymore. After all, it is about 7-6 weeks old. A quarter to half a pack of dry yeast is the easiest way at this point. It also doesn't matter if ale or lager yeast is used since the flavor profile of the beer has already been determined by the yeast used for the primary fermentation. If you don't add fresh yeast you need to be more patient with the conditioning of the beer. Let the beer carbonate at room temperature or anywhere between. The higher the temperature is, the faster the beer will carbonate.

The following sections will explain lager fermentation in more detail and show procedures that can improve the quality of the finished beer even further.

The 3 phases of a lager fermentation

A lager fermentation consists of these 3 phases:

- primary fermentation : main fermentation of the fermentable extract. The bulk of the CO₂ and alcohol are created here
- maturation : the yeast is allowed to clean up some of its byproducts like diacetyl (butterscotch flavor) and acetaldehyde (green apple flavor)
- cold stabilization (lagering) : the low temperature causes haze forming proteins and polyphenols come out of solution and drop out of suspension. There is also a mellowing of flavors and some formation of esters happening. The latter

becomes only significant after more than 12 weeks [Narziss 2005]

Depending on the fermentation schedule that is used, which are explained in the following sections, distinct boundaries may exist between these phases or they may simply flow into each other. In particular the maturation of the beer (also known as diacetyl rest) can be held at a temperature higher than primary fermentation, an extension of primary fermentation or at a temperature lower than primary fermentation.

The conventional fermentation in a German lager brewery

The majority of the information given in this section is taken from a German brewing text book "Abriss der Bierbrauerei" (Overview of beer brewing) by Ludwig Narziss, one of Germany's leading teachers and experts on brewing.

After the whirlpool the wort is cooled to close to 32 °F (0 °C) to maximize the cold break. It is then warmed up to pitching temperature which can be between 41 °F (5 °C) and 46 °F (8 °C). The majority of the cold break (~ 60%) is removed through either sedimentation tanks, flotation tanks, centrifuges or filtration. Filtration is the only means of complete cold break removal the other methods remove only about 2/3 of it. Once the cold break is removed the wort is aerated with sterile air to achieve a wort oxygen content of 8-10 ppm (mg/l). In case of cold break removal through flotation, the aeration of the wort is achieved during the flotation process.

Yeast is pitched at about 500 ml thick yeast slurry per 100 l 12 °P (1.048 SG) wort (this equals about 100ml or 3oz yeast slurry per 5 gal). Once the yeast is well distributed this equals about 15×10^6 cells per ml wort. When it comes to pitching and primary fermentation temperatures cold and warm lager fermentation exists. The cold fermentation uses a pitching temperature of 41 °F (5 °C) and a maximum fermentation temperature of 48 °F (9 °C) and the warm fermentation uses a pitching temperature of 46 °F (8 °C) and a maximum fermentation temperature of 50 - 54 °F (10 - 12 °C). This should however not be confused with warm vs. cold pitching. There is no warm pitching in commercial German lager fermentation.

Once the yeast is pitched it takes about 24 hours for the low Kraeusen to develop. High Kraeusen starts on the 3rd day, when the maximum temperature is reached and lasts until the 5th day. At this time yeast growth slows down and the yeast starts to flocculate. This is when the beer is slowly cooled at a rate of 0.5 - 0.7 °C in order to avoid shocking the yeast. At this time the primary fermentation is considered done, but the beer has only attenuated to about 40% - 60%.

A few days later, when the beer is racked to the secondary/lagering tanks it has a temperature of about 39 - 41 °F (3.5 - 5 °C). The remaining fermentable extract is 1.2 - 1.4 % by weight (about 5 - 6 gravity points). Many breweries mix beer from different batches in the lagering tanks to compensate for fluctuations in color, bitterness, attenuation and other parameters. The addition to that 20-50% beer fermented with low flocculating yeast is beneficial for achieving a higher attenuation since this yeast will work longer than better flocculating strains of yeast.

During the secondary fermentation (a.k.a lagering) the tanks are closed and the pressure build-up is controlled by a pressure sensitive bleeder valve. This system, called Spundungsapparat, ensures the proper carbonation of the beer during lagering. The German Purity Law prohibits the use of non-fermentation CO₂ for beer carbonization. It is also more economical for a brewery to use the CO₂ produced during fermentation.

Key for a good lagering is control of the yeast contents and temperature profile such that the fermentation slowly continues during the whole time the beer is lagered. Only this allows for the processes to happen that are commonly referred to as maturation: reduction of diacetyl, acetaldehyde, higher alcohols etc. The lagering takes between 4 weeks and 6 months. At the end of lagering the beer has the desired attenuation, which is generally a little higher than the limit of attenuation. For light colored beers this attenuation is about 2-4% and for dark beers as much as 6% above the limit of attenuation. Export style beers can have an attenuation as close as 0.5% below the limit of attenuation. A difference between actual and limit of attenuation means that there are fermentable sugars left in the beer which are a vital part of the flavor profile, but larger percentages of these sugars result in a less shelf stable beer.

Brewing lagers in a home brew setting

The process described above is how the best lagers are brewed. But it is very difficult to handle for the home brewer. The reason: the yeast needs to be kept working throughout the lagering phase otherwise the result is an underattenuated (=sweet) beer that is rather undrinkable and requires further attention (mostly the addition of fresh yeast and raising its temperature).

Because of that all home brewing instructions that are given in the state of the art home brewing literature are based on a lager fermentation process that is referred to as accelerated fermentation and maturation in commercial brewing. Even many commercial lager breweries use this accelerated process due to time and tank space constraints for the production of their lager beers. Anheuser Bush for instance produces Budweiser with only one week of primary fermentation and 3 weeks of secondary/lagering. The key to this is their Beechwood ageing process where the porosity of the beechwood allows for a greater contact area between the yeast, which flocculated onto the beechwood strips, and the beer resulting in a shorter maturation time.

Home brewing a lager (the advanced process)

The following sections go into more detail about the home brewing process for lagers and try to explain the different approaches to the process that are done by different brewers.

Pitching rate and yeast propagation

Proper pitching rate is important for brewing lagers, especially if cold pitching (see next section) is chosen. As noted previously the proper amount of yeast for a 12 °P (1.048 SG) wort is about 100 ml yeast sediment for 20l (5 Gal) of wort. Jamil Zainasheff from the Brewing Network has a nice pitching rate calculator (<http://www.mrmalty.com/calc/calc.html>) on his web page. This tool allows you to calculate the amount needed for proper ale and lager fermentation based on the original gravity of the wort and the viability of the yeast.

million cells per Plato and ml wort		1.25
ml sediment per Plato and liter		0.42
billion cells per ml sediment		3

wort strength		yeast needed for pitching			
Plato	SG	million cells per ml wort	ml sediment per liter wort	ml sediment per 20l wort	oz sediment per 5 gal wort
10	1040	13	4.2	83	2.6
12	1048	15	5.0	100	3.2
14	1056	18	5.8	117	3.7
16	1064	20	6.7	133	4.2
18	1072	23	7.5	150	4.7
20	1080	25	8.3	167	5.3
22	1088	28	9.2	183	5.8

Lager yeast pitching amount based on the recommendation of 100 ml thick yeast sediment per 100l 12 Plato wort [Narziss 2005]

The problem with pitching based on a pitching rate is always knowing how much yeast one actually has available. Without a Hemocytometer (<http://en.wikipedia.org/wiki/Hemocytometer>) it is hard to count yeast cells. That's why determining yeast amount based on the volume of yeast sediment is more practical for the home brewer.

The pitchable liquid yeast products that are out there (White Labs vials and Wyeast Activator packs) are not a sufficient amount of yeast for lagers when pitched cold. This means you need to propagate yeast by growing them in wort. Essentially you are making a starter but you are expecting more than waking up the yeast; you are expecting yeast growth. That's why the starter needs to be fairly big and needs to be prepared a few days in advance. How big depends on your equipment and how early depends on the temperature at which you keep the starter.

Equipment: If you have a stir plate you should be fine with about 2 qts of 10 °P (1.040 SG) wort because the constant yeast suspension and aeration leads to a more efficient yeast growth. If you don't have a stir plate and need to shake the starter once in a while to keep as much of the yeast suspended as possible, you should make a 3-4 qt starter. These amounts assume that you want to grow yeast from a White Labs vial or a Wyeast activator pack. Experience will tell you what works best for you.

Temperature: As much as there is debate about warm pitching vs. cold pitching there is debate about the propagation temperature for lager yeast. Microbiologists will tell you that yeast (lager and ale) should be propagated at 75 °F (25 °C) because that is the temperature at which they will grow best. Brewers however generally agree that yeast should be propagated at or slightly above primary fermentation temperature because the yeast should not get used to living and performing at higher temperatures than the primary fermentation temperature. Some home brewers report that yeast grown at temperatures above room temperature (80+ °F) tends to lose their flocculation characteristic which makes it harder for the yeast to drop out at the end of fermentation. Growing yeast at primary fermentation temperatures also avoids shocking the yeast when pitched because its temperature is already close to pitching temperature. The yeast propagation guidelines (<http://www.hefebank-weihenstephan.de/page/Downloads/propagation-English.pdf>) from the yeast bank Weihenstephan also suggest that the last stages of propagation are done close to fermentation temperatures.

Reusing an existing yeast cake is the easiest way to get a proper pitch of yeast. But it is recommended that the wort is not simply racked onto the old cake but that the yeast is taken from the primary fermenter and pitched according to the necessary pitching rate for the beer to avoid over pitching. Keep in mind that yeast quickly loses its vitality after primary fermentation and that such yeast should be stored cold (close to 32 °F is best) and pitched within a week. After that the yeast performance starts to suffer significantly and it might be necessary to make another propagation step with that yeast to recharge its glycogen reserves.

Cold vs. Warm pitching

Among home brewers there is often debate regarding the proper pitching temperature for lagers. Some say that you need to pitch warm to allow for better initial growth of the yeast and others say that you need to pitch below the primary fermentation temperature.

To understand that both sides have valid arguments one has to understand where they are coming from. Warm pitching has been introduced by home brewers and yeast manufacturers because it allows for pitching a lager with a smaller pitching rate and leads to a shorter lag time which is less concerning for the first time lager brewer. That's why warm pitching was suggested in the "Your first lager fermentation" section above. To pitch warm, chill your wort until it has a temperature of 65 - 68 °F (15 - 18 °C) aerate it well and pitch the yeast. Now wait until you see signs of fermentation (low krausen or bubbles in the airlock) and move it to an area where you maintain about 50 °F so that the wort can cool down while the yeast starts to take off.

Industrial lager brewing only does cold pitching because the proper pitching rates and yeast health can be ensured. Because of the initially higher fermentation temperatures, warm pitching is associated with an increased level of ester, diacetyl and fusel alcohol production which are components that, at higher levels, are not desired in a lager. Diacetyl will be reduced by the yeast during the maturation of the beer but most of the esters and higher alcohols can carry over into the finished product.

That's why cold pitching is also recommended for the home brewing of lager beer as long as a proper pitch of healthy yeast is available. If this is not the case, cold pitching can lead to a very long lag time and sluggish fermentation. In order to pitch cold, chill the wort to 43 - 48 °F (6 - 9 °C) and resuspend the yeast with some wort. Make sure to break up all clumps. If you have a stir plate, stirring the wort and yeast for a few minutes will take care of that very nicely. Now pitch it into the wort and place the fermenter in a space with a constant temperature of 46 - 50 °F (8 - 10 °C). Most brewers use a fridge or freezer chest with an external temperature control for this. Expect the fermentation to start within 16 - 36 hrs. For lager fermentation the lag time should not be too short. Because of the low beer (it is actually beer since yeast has already been pitched) temperature there is less risk of contamination since the metabolism of the bacteria and wild yeast is also reduced. The lag time also appears longer for lagers because the colder beer can absorb more CO₂ before it is forced out of solution and forms the Krausen. If you are concerned that your yeast isn't active after pitching, measure the pH of the beer. If it dropped from the lower to mid 5's at pitching time into the upper 4's 12 hrs after pitching, the yeast is doing alright. The lowered pH already provides additional insurance against infections. Oftentimes you can also see a stratification of yeast. In this case the very top layer of the beer seems darker and less cloudy. The yeast is in suspension in the cloudy beer just below that layer. I have had low Krausen formation as late as 48 hrs after pitching. Though I don't like it to take that long, the beer turned out just fine.

Fast Ferment Test

When brewing lagers I strongly recommend that a Fast Ferment Test is done. Palmer [Palmer, 2006] was referring to brewing lagers as flying blind because you don't know when they are done fermenting. With a FFT you will be able to determine the FG of your lager well before the actual batch has finished fermenting. This is especially helpful for all grain brewers because the final gravity of the beer can greatly be influenced by mashing.

Primary Fermentation

Because of the slower yeast metabolism at lower fermentation temperatures, lager fermentations take longer than ale fermentations of the same wort. While ale primary fermentations are generally done after 3 - 6 days and final gravity is reached at that time, for lagers it can take 1 - 3 weeks and the final gravity may not be reached after the completed primary fermentation.

In the classical lager brewing method, as described above, the primary fermentation is over after about 7 - 10 days, but the attenuation of the beer is not yet at the attenuation level that is desired at bottling time. Good fermentation management allows the yeast to be actively fermenting even during the lagering (cold storage) phase. This needs to be kept in mind when brewers talk about the length of primary fermentation for their lagers: What was the attenuation when the beer was racked to a secondary and what was the attenuation of the beer when it was done?

Proper temperature control is crucial for a clean lager fermentation. This can be done by controlling the temperature of the beer or the ambient temperature. If you control the temperature of the beer, you are able to avoid a temperature drop once the fermentation slows down. This will keep the yeast more active and the maturation will progress faster. But it is not really necessary to do so. Keeping the ambient temperature at a constant level is sufficient if this temperature is chosen such that the beer will not exceed a maximum fermentation temperature of 46 - 54 °F (8 - 12 °C) which is usually reached shortly after the high krausen stage. Lower temperatures will cause longer fermentation times, but are known to yield better beers due to the further suppression of fusel alcohol formation. Because of the slower yeast metabolism at these low temperatures, the fermentation temperature is not expected to rise more than 2-4 °F (1-2 °C) above the ambient temperature which means that there won't be a significant temperature drop once the fermentation slows down. The difference between the ambient temp and the fermentation temperature is also a nice indication for the intensity of the fermentation.

Though many lager yeast strains indicate that they ferment with lager characteristics even in the upper 50's (14 °C) I recommend that the primary fermentation temperature shouldn't exceed 54 °F (12 °C). Keeping it closer to 48 °F (9 °C) is even better.

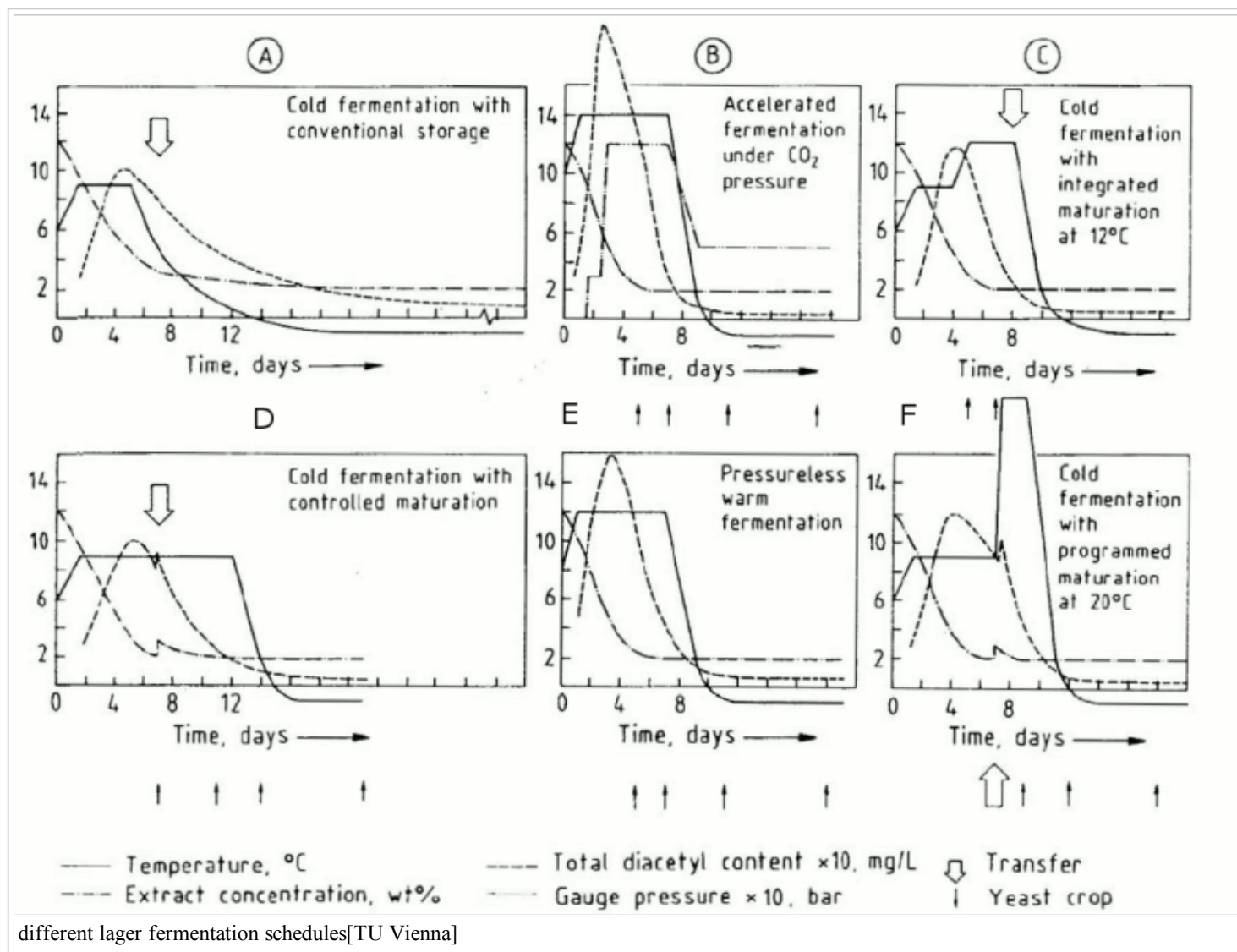
Once you see the activity of the fermentation slowing down significantly you should take a gravity reading of the beer to check its current attenuation as well as taste. Once you have brewed a few batches of lager beer, this attenuation will also give you an idea of the yeast's performance during the primary fermentation. Keep a record of this to compare different yeast strains and other fermentation factors on the fermentation performance.

Maturation of the beer

Once the primary fermentation is considered done the final gravity has not been reached yet and fermentation byproducts like diacetyl and acetaldehyde need to be reduced by the yeast. This process is called maturation of the beer and in the conventional fermentation approach for lagers, as outlined above, it happens during the long cold storage.

But most of the books and instructions on home brewing lagers don't suggest this fermentation practice because it is very easy to fail. The problem lies in not being able to chill the fermenting beer without shocking the yeast into dormancy. The latter will result in a cloyingly sweet and under attenuated beer that may still have elevated levels of diacetyl because the yeast was not able to sufficiently reduce these "young-beer" compounds. The key to such a fermentation schedule are excellent yeast health, precise temperature control and a yeast strain that works well at very low temperatures. Many of these factors are much easier to achieve in a commercial brewery than in a home brewery.

To avoid this problem, home brewers usually use what is referred to as accelerated maturation in commercial brewing, a process which allows for the nearly complete fermentation of the beer before it is placed into cold storage. To further understand the various maturation techniques that are out there let's have a look at the following diagrams which I found in a presentation about beer production on the web server of the Technical University of Vienna [TU Vienna]:



different lager fermentation schedules[TU Vienna]

(A) - shows the temperature (continuous line), extract (=gravity) (dash-dot-dash line) and diacetyl (dashed line) of a conventional lager fermentation. As you can see the maximum fermentation temperature is held for only 4 days before the beer is slowly cooled to lagering temperatures over the course of the next 7 to 8 days. Though the extract is close to the FG of 2 °P, the diacetyl level is still considerable at the time of racking (large arrow on top). This diacetyl is reduced over the course of lagering.

(B) - shows a lager fermentation schedule that uses higher fermentation temperatures and pressure to accelerate the primary fermentation and maturation of the beer. The controlled use of pressure (indicated as dash-dot-dot-dash line in the diagram) allows the production of esters and higher alcohols, which is more intensive at higher fermentation temperatures, to be limited [Narziss, 2005]. Though more diacetyl is produced because of the higher fermentation temperature it is also reduced quicker, allowing the beer to mature within only 8 days. primary fermentation under pressure is for the home brewer only of academical interest (though some have done it successfully) since it requires primary fermenters that can withstand pressure and a means of controlling that pressure without the benefit of an increase in beer quality.

(C) - This fermentation schedule comes closer to what a home brewer can do. The wort is pitched cold at 44 °F (6 °C) and rises to 48 °F (9 °C) over the next few days where it is kept until an attenuation of 40 - 50% is reached. The fermentation temperature is then raised to give the fermentation another boost which results in reaching the beers final gravity and the reduction of diacetyl below the taste threshold. This step is commonly referred to as the "diacetyl rest" though the temperature increase is not as pronounced as the one shown in diagram F. After that rest, the beer is racked from the yeast and quickly chilled to lagering temperatures. Since no further yeast activity is necessary (target attenuation has been reached and diacetyl has been reduced) there is no need in a gradual cooling of the beer to avoid shocking the yeast. This works well if you have a dedicated lagering space in which you keep multiple batches and thus need to keep its temperature constant.

D - Here is another fermentation schedule better suited for the home brewer than (A). The wort is pitched cold and primary fermentation is done at 48 °F (9 °C). Once the beer is within about 2 °P (8 gravity points) of the targeted FG (see Fast Ferment Test) the beer is racked to a secondary. Because of the colder fermentation temperature the addition of kraeusen beer can be beneficial for a better diacetyl reduction as well as better attenuation of the beer. For an explanation of that process see below. If you use a soda keg as the secondary fermentation vessel (make sure you shorten its dip tube by about one inch) the secondary fermentation can be used to carbonate the beer. Besides having a few practical reasons (keeping the keg sealed and having carbonated beer once the lagering is complete) it takes the home brewing process closer to the way lagers are brewed in Germany. Since the German Purity Law for beer doesn't allow the use of non-fermentation CO₂ for carbonation, brewers need to carbonate their beer during the secondary fermentation and lagering phase. Once the beer has been racked to a carboy or keg, it is kept at primary fermentation temperature until the targeted attenuation has been reached and the diacetyl has been sufficiently reduced. After that the beer can quickly be chilled to lagering temperatures because there is no need to keep the yeast working.

Another widely used approach is to keep the beer in the primary fermenter until it has completely fermented and the diacetyl has been reduced. While it is not as easy to build up carbonation with this process, the larger amount of yeast available in the primary allows for a quicker maturation of the beer.

E - This is similar to D, but the pitching and primary fermentation temperatures are higher which results in faster fermentation and maturation.

F - Is a fermentation schedule that uses an explicit maturation rest at a higher temperature: also known as diacetyl rest. This schedule resembles best what authors like Noonan and Palmer suggest for a proper lager fermentation: Pitch cold, let it ferment around 50 °F (10 °C) and once the fermentation slowed down significantly and the gravity of the beer is close to its final gravity, raise the beer's temperature to 65 - 68 °F (17-19 °C) for a diacetyl rest. This diacetyl rest has the effect of giving the slowing yeast a boost to finish the last sugars and reduce the diacetyl. As the previous examples for fermentation schedules showed, such a rest is not really necessary. But it can be helpful when the used yeast shows a very sluggish fermentation performance and has a hard time reaching the targeted final gravity when kept at fermentation temperatures. After this rest is complete the beer can be crashed to lagering temperatures.

These 6 examples, which were taken from a technical brewing and not a home brewing text, illustrate the various options for lager fermentation control that the home brewer has. A very practical option is D since it requires only 2 fermentation spaces at constant temperatures: one at 46 - 50 °F (8 - 10 °C) for the primary fermentation and maturation and one at 32 - 36 °F (0 - 2 °C) for the cold conditioning/lagering. Since no ramping of temperature is necessary such a fermentation cellar set-up easily allows for the fermentation/lagering of multiple batches of lager at different stages. Since the actual temperature of an explicit maturation rest (diacetyl rest) is not so important it can easily be accomplished by moving the beer into a warmer part of the house before starting to cold condition (lager) the beer if such a rest is needed.

When to rack the beer

When using accelerated maturation, as described above, the beer can be racked before or after its maturation is completed. Though it will be taken from the majority of the yeast, plenty of yeast will remain in suspension to finish the job although it may do its job a little slower. My own experience tells me to keep the beer in the primary until it is at least within 1 Plato (4 gravity points) of the expected final extract/gravity. Racking shortly after finished primary fermentation also enables the home brewer to harvest fresher yeast that can immediately be used in another batch.

Adding Kraeusen

Kraeusen beer is fermenting wort which is still in its high kraeusen stage. When added to a beer after primary fermentation, it provides fresh healthy yeast, which may be able to do a better job in attenuating and maturing the beer. Sometimes it can also be beneficial to add different yeast with the addition of Kraeusen. A less flocculent strain, for example, can enable a better attenuation while the main flavor profile was determined by the more flocculent strain that was used during primary fermentation. For a commercial brewer the addition of Kraeusen is simple since they always have various batches of the same beer at different fermentation stages. Some of them may even use different yeast strains for the benefit mentioned above.

In order to use the kraeusen technique, the home brewer will have to make a large (1 - 2 l or quart) starter from either fresh wort or wort saved from the initial batch pitched with some of the yeast from the primary, or a new culture of yeast. Left over wort from brewday can easily be kept in soda bottles in the freezer if it is boiled and cooled before the yeast is pitched (*I have*

to add a wort of caution here as I'm currently suspecting that oxygenation of the wort is occurring during that time and it eventually leads to an off-flavor of the beer. Until I determined the exact source of that off-flavor I'm not using wort stored in the freezer anymore. But the wort stored in the freezer is still great for propagating yeast). One way of getting some yeast out of the primary is to use a sanitized racking cane. Use your thumb to keep one end closed and push it into the carboy. Then release the thumb and beer will rush into the cane and pull a lot of yeast with it. Close the cane off again, pull it out and dump its contents in the starter vessel. This should be repeated 5-10 times to get enough yeast into the starter. Now the yeast in the starter is allowed to start fermentation at primary fermentation temperatures. When the beer is then racked to the secondary, the Kaeusen is added without adding the layer of yeast sediment that may have already settled on the bottom.

Maturation/Cold Conditioning Vessel

There are 2 kinds of vessels that are commonly used by home brewers for the maturation and cold conditioning (a.k.a. lagering) of lagers: carboys and soda kegs. Because of their oxygen permeability, buckets should not be used for long term storage of any beer (except some wild beers) and are thus not suitable for lagering. When the beer is racked after the fermentation and maturation has been completed, the carboy or keg should be purged with CO₂ to minimize the amount of oxygen that is left in there once the beer has been transferred because the inactive yeast will not be able to bind this oxygen. Though more expensive, soda kegs make for the best lagering vessel for the home brewer:

- Their narrower shape allows more of them to fit in a freezer chest or fridge thus allowing for a more efficient use of the cold space
- The beer can be carbonated (force or natural) during the lagering phase. This may actually be necessary to keep the lid sealed
- With a shortened dip tube the sediment can be left behind and the beer can easily be transferred to a serving keg by using pressure or siphon without any contact to the ambient air.
- They don't break

Natural carbonation

Unlike commercial German brewers, home brewers are not required to carbonate their beers naturally. But using natural carbonation provides a few benefits that should not be overlooked:

- Since the yeast is still fairly active when the beer is racked, most of the oxygen that is picked up during this process (splashing or O₂ in the head space) will be taken up by the yeast before it can react with other compounds in the beer. This is beneficial because due to the shortened dip tube the lagering keg cannot be purged with CO₂ as efficiently as serving kegs can. To efficiently purge a serving keg (full length dip tube) fill that keg completely with sanitizer solution and push it out with CO₂ and don't open the keg after that. Because there was no air in the keg when it was closed off, all the sanitizer has been replaced with CO₂. Obviously this doesn't work so well with a shortened dip tube since a considerable amount of sanitizer would remain in the keg. Recent studies have shown that even active yeast may not be active enough to consume all the oxygen from the headspace before it can contribute to oxygenation of the beer [Hermann 2005], because of that the head space should be purged as well. It is sufficient to vent and re-pressurize it for 4-5 times.
- When the keg is pressurized by the CO₂ from the fermenting beer, there is no connection to a CO₂ tank necessary to keep the lid sealed and to take samples with a picnic tap
- The beer is already carbonated when the lagering is completed.

To use natural carbonation a soda keg or other pressure resistant vessel needs to be used. Carboys are not designed to withstand pressure. Since CO₂ needs to be produced for the natural carbonation a sufficient level of fermentable sugars needs to be present after the beer has been racked. This can be achieved by:

- racking the beer when about 1-1.5 % fermentable extract (4 - 6 gravity points) are left. At this time there is also enough yeast in suspension to ensure a good secondary fermentation. This process is called Gruenschlauchen (green racking) in German brewing
- adding fermentable sugars in the form of Speise (a.k.a gyle, which is unfermented wort), malt extract or sugar. But if the beer has already fermented too far the amount of yeast that is transferred during racking may not be sufficient for a good secondary fermentation

- adding Kraeusen allows the addition of fermentable extract and fresh yeast. This is the preferred method to use when the beer is racked late and the amount of yeast in the racked beer is lower. It also allows topping off the keg with another 2 qts of beer when the primary fermentation was done in a 5 gal carboy and only 4.5 gal beer could be racked to the keg.

Contrary to priming bottles, the amount of sugar needed for natural carbonation doesn't have to be calculated as precisely since the actual carbonation in the beer can easily be adjusted with Spunding. Spunding is the technical German term for controlled pressure release from a fermenter. For this a pressure gauge needs to be connected to the gas-in connect of the keg. Based on a carbonation table the current CO₂ content of the beer can be determined from the current head pressure (measured by the pressure gauge) and the current temperature of the beer. If the carbonation is too high, the excess pressure can be relieved with the bleeder valve. If the bleeder valve is built as an adjustable pressure sensitive blow-off valve the process is even easier since you can set the maximum pressure based on the desired carbonation level for the given beer temperature and let the system vent fermentation CO₂ as needed.

lagering/cold conditioning

Whereas maturation and cold conditioning happen simultaneously with the classic lager fermentation schedule, it is a separate step when accelerated maturation is used. With that the beer has been almost completely fermented before the temperature is lowered to near freezing and no particular care needs to be taken to avoid shocking the yeast since no significant further yeast activity is required. When the beer is conditioned at low temperatures various processes take place that lead to the smooth character which is expected from a lager:

- Proteins and polyphenols (tannins) form agglomerations (basically bind with each other to form larger molecules) which become insoluble and precipitate out of solution. [Nguyen 2007]
- Hop polyphenols will drop out leading to milder hop bitterness
- Yeast sediment which cleans up the beer and removes the yeasty smell and taste associated with young beer
- Some of the alcohols and acids form esters in the beer which leads to new flavor compounds. This process is very slow and becomes only significant after more than 12 weeks [Narziss 2005].
- Some yeast activity may be present which leads to further clean-up and extract reduction of the beer. I oftentimes see another extract drop of 0.1 - 0.2 Plato over the course of a few weeks.

Depending on the gravity and style of beer, a cold conditioning time of 4 weeks up to 6 months is required. Towards the end of that time, fining agents like gelatin or isinglass might be used to speed up the clarification process, but are generally not necessary. Once the beer has finished cold conditioning, it should be racked to a serving keg to get it off the old yeast and avoid autolysis of this yeast. At this point it is also safe to let the beer warm up again

racking to a serving keg

If a soda keg was used for cold conditioning the transfer can be done in a closed system which dramatically reduces the beer's exposure to oxygen, which at this stage is the beer's biggest enemy due to the unavailability of active yeast that could absorb it. When attempting to rack in a closed system, the shortened dip tube is important. It lets you move the beer and leave the majority of the sediment behind. The trick is to move the cold conditioning keg as little as possible before the beer is racked because any movement can stir up sediment. If you are using a freezer chest to cold condition the beer you can leave the keg in the freezer chest while using CO₂ to push it into the serving keg. If the beer is already carbonated you need to mount a pressure gauge and valve on the gas-in port of the destination keg to maintain pressure in that keg. Otherwise the beer will foam up. You can then open the valve a little to let some of the CO₂ escape while the beer is flowing into that keg. Keep an eye on the kegs when the destination keg is almost full. Once CO₂ is blown through the jumper hose, the beer in the destination keg can foam up and enter the pressure gauge/valve.



soda keg with pressure gauge and bleeder valve

Another efficient and more elegant way of racking beer from one keg to another is to siphon the beer from one keg to another. For that place the source keg above the destination keg. Connect the gas-in of the destination keg and the gas-in of the source keg to the same CO₂ regulator. This results in the same pressure in the destination and source keg. Now take a long jumper hose and first connect it to the bev-out of the destination keg and then to the bev-out of the source keg. Always make sure to connect to the bev-out of the source keg last. If there is pressure in the jumper hose, you could bubble CO₂ through the dip tube on the source keg which will stir up sediment.

Nothing is flowing yet. To get the siphon started disconnect the CO₂ on the destination keg and briefly pull the pressure relieve valve on that keg. Now that the pressure is slightly lower in the destination keg the beer will start flowing and the siphon is started. Reconnect the gas-in of the destination keg to the CO₂ system so the CO₂ from that keg can flow into the source keg. This process will take a while, but you don't need to monitor it.

If you plan to filter your beer, the transfer from the cold conditioning keg to the serving keg is the time to do this.

Instead of racking the beer to a serving keg it can also be filled into bottles at this stage. The cold temperature of the beer allows it to retain its carbonation much easier. So well in fact that it can be filled into bottled chilled to beer temperature without the use of a counter pressure bottle filler. A piece of racking cane stuck into the tip of a picnic tap helps with filling the bottles from the bottom up. Cap on foam to minimize trapping air in the bottle. I do that for all lagers that I plan to age in bottles for an extended time after they have been cold conditioned.

lagers and bottle conditioning

When bottle conditioning lager beer, there are 2 options for the brewer:

- bottle the beer after the primary fermentation and maturation is complete and cold condition the beer in the bottle
- bottle the beer after cold conditioning is complete

Both have their pros and cons

bottle conditioning before cold conditioning

When you bottle condition before cold conditioning, you wait until the beer has completed fermentation and prime the beer with corn sugar or DME. Since the yeast is still fairly healthy and active there shouldn't be any problems in getting the beer carbonated. Let the beer carbonate at room temperature for a week. Give it a taste to ensure complete carbonation before moving it to cold storage 32 - 42 °F (0 - 5 °C) to cold condition it.

Because the beer is bottled before cold conditioning, all the yeast and other sediment that settles out during that phase will remain in the bottle.

bottle conditioning after cold conditioning

If you plan to bottle after cold conditioning, as suggested in Noonan's Book "New Brewing Lager Beer", you lager the beer in a carboy first (since you are bottling I'm assuming that you don't have kegs for cold conditioning). This may take 4 weeks to a few months depending on the beer. Because the yeast has been inactive for such a long time and only little yeast is in suspension anyway, it is recommended to add fresh yeast at bottling time to ensure consistent carbonation in a reasonable time frame. The fresh yeast can come from either dry yeast (1/4 pack should be enough), yeast sediment from the primary fermenter of another batch or Kraeusen. The type of yeast doesn't matter much since the flavor has already been defined during the fermentation and cold conditioning process. Any clean well flocculating ale or lager yeast will do.

When racking from the cold conditioning vessel to the bottling bucket make sure that as little sediment as possible is transferred since the advantage of this method is to leave all this behind and have the beer benefit from bulk-aging.

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